SIMATS ENGINEERING
ASSESSMENT TOOL 3: Mock Interview

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA1522

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Name	**192525076**

COURSE OUTCOME COVERED
CO5: *Design big data processing systems using the Hadoop ecosystem including HDFS, MapReduce, and YARN.* (BL5)

Dave's Taxonomy: Articulation (Level 4) **Total Marks: 25**
Question Count: 10

Code of Conduct
I **R.Pranathi(192525076)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: R. Pranathi

Assessment Tasks
Mock Interview
Answer the following interview-oriented questions clearly and concisely. Support your responses with architecture-level reasoning wherever appropriate.

1. Explain the difference between HDFS and traditional file systems.
2. Why is replication important in distributed storage systems?
3. How does MapReduce divide work between map and reduce phases?
4. What role does YARN play in large-scale Hadoop processing?
5. Differentiate NAS and SAN in an enterprise data environment.
6. What is the purpose of ETL in a big data pipeline?
7. How does Apache NiFi help in data integration workflows?
8. What are the key failure points in a Hadoop cluster and how would you handle them?
9. How would you design a secure data ingestion pipeline for a large organization?
10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

Mock Interview Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Knowledge	25%	Shows deep and accurate technical understanding	Good technical understanding	Basic understanding	Weak understanding
Problem Solving	20%	Responds with strong architecture-level reasoning	Reasonable reasoning	Basic reasoning	Weak reasoning
Communication	20%	Answers are confident, precise, and professional	Mostly clear communication	Moderate clarity	Weak communication
Practical Awareness	20%	Connects concepts to real-world implementation well	Some practical relevance	Limited practical relevance	No practical linkage
Confidence and Professionalism	15%	Highly confident and professional	Generally professional	Moderately professional	Low confidence and professionalism

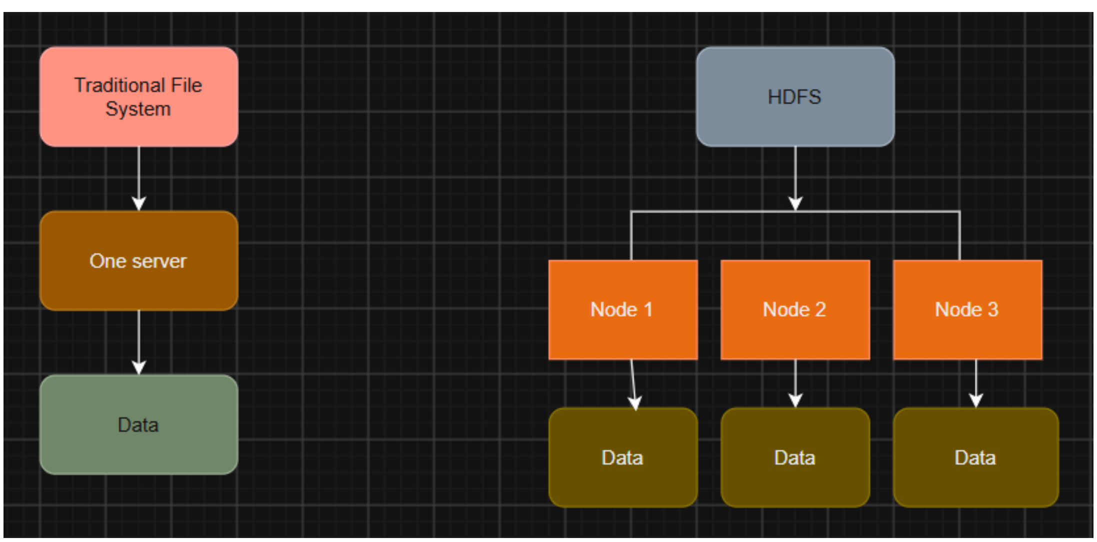
1. Explain the difference between HDFS and traditional file systems.

Ans:

HDFS (Hadoop Distributed File System) is designed to store very large amounts of data across multiple computers, while a traditional file system usually stores data on a single computer or storage server.

In a traditional file system, if the storage server fails, the data may become unavailable. In HDFS, a large file is divided into blocks and these blocks are stored across multiple DataNodes. HDFS also keeps multiple copies of the blocks using replication, so the data can still be accessed if one node fails.

For example:



HDFS is therefore suitable for big data because it provides distributed storage, scalability and fault tolerance. Traditional file systems are more suitable for normal applications where data does not need to be distributed across many machines.

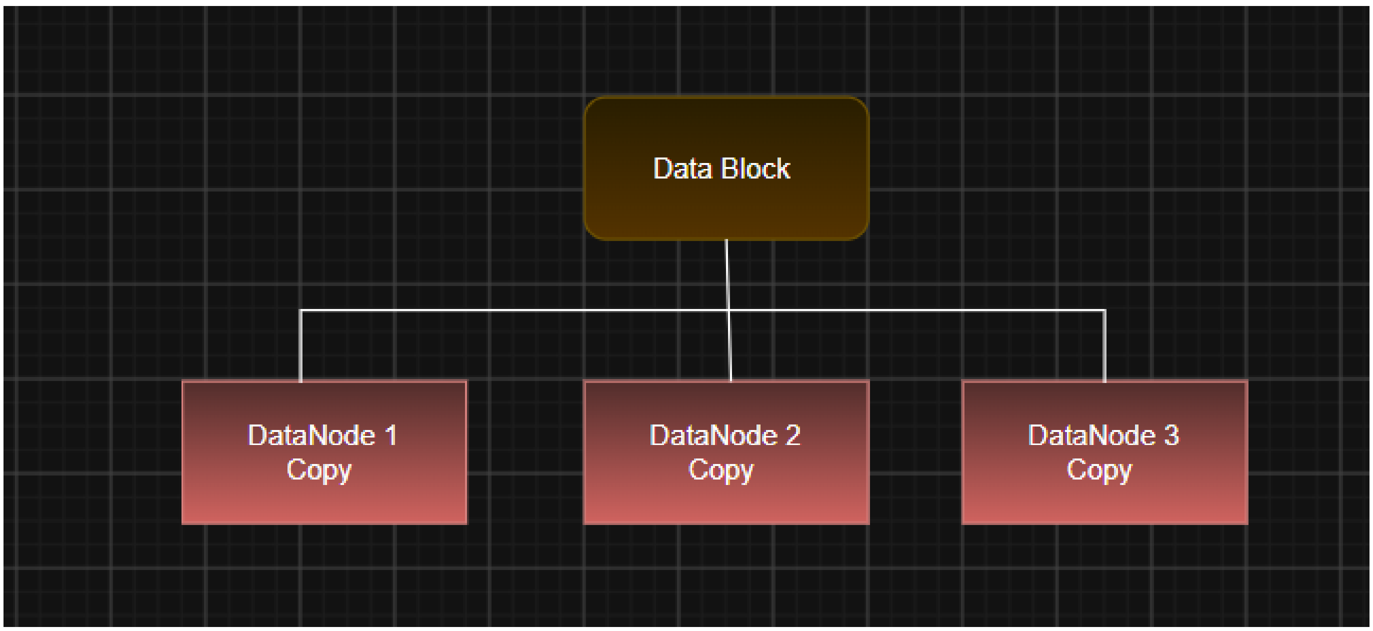
2. Why is replication important in distributed storage systems?

Ans:

Replication is the process of **keeping multiple copies of the same data on different machines** in a distributed storage system.

It is important mainly for **data availability and fault tolerance**. If one server fails, another server can provide the same copy, so the data is not lost or unavailable.

For example, in HDFS, a data block can have **three replicas**:



If **Node 1 fails**, the system can read the data from Node 2 or Node 3.

Replication also helps during hardware failures and improves system reliability. However, keeping more copies requires more storage space and increases cost. Therefore, the replication factor should be selected based on the required reliability.

3. How does MapReduce divide work between map and reduce phases?

Ans:

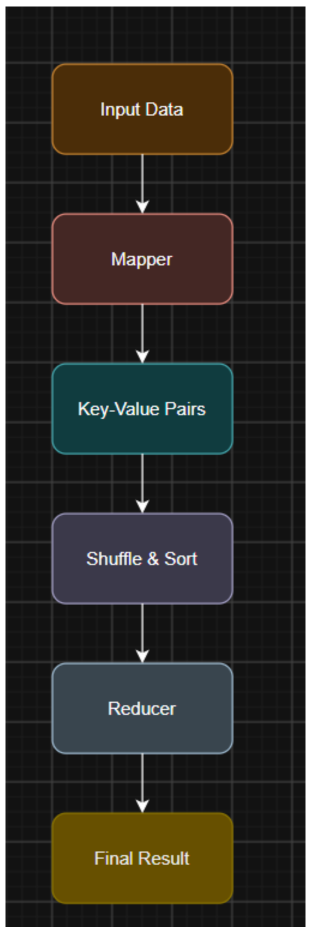
MapReduce divides a large data-processing task into two main phases: **Map and Reduce**.

In the **Map phase**, the input data is divided into smaller parts, and multiple Mapper tasks process these parts in parallel. Each Mapper produces **key-value pairs**.

Then, during the **Shuffle and Sort phase**, the intermediate data is grouped based on the keys.

In the **Reduce phase**, Reducer tasks process the grouped data and produce the final result.

For example, in word counting:



For example:

"cat dog cat"

Mapper → (cat,1) (dog,1) (cat,1)

Reducer → (cat,2) (dog,1)

So, **Map divides and processes the input data, while Reduce combines and summarizes the intermediate results**. This allows large datasets to be processed in parallel.

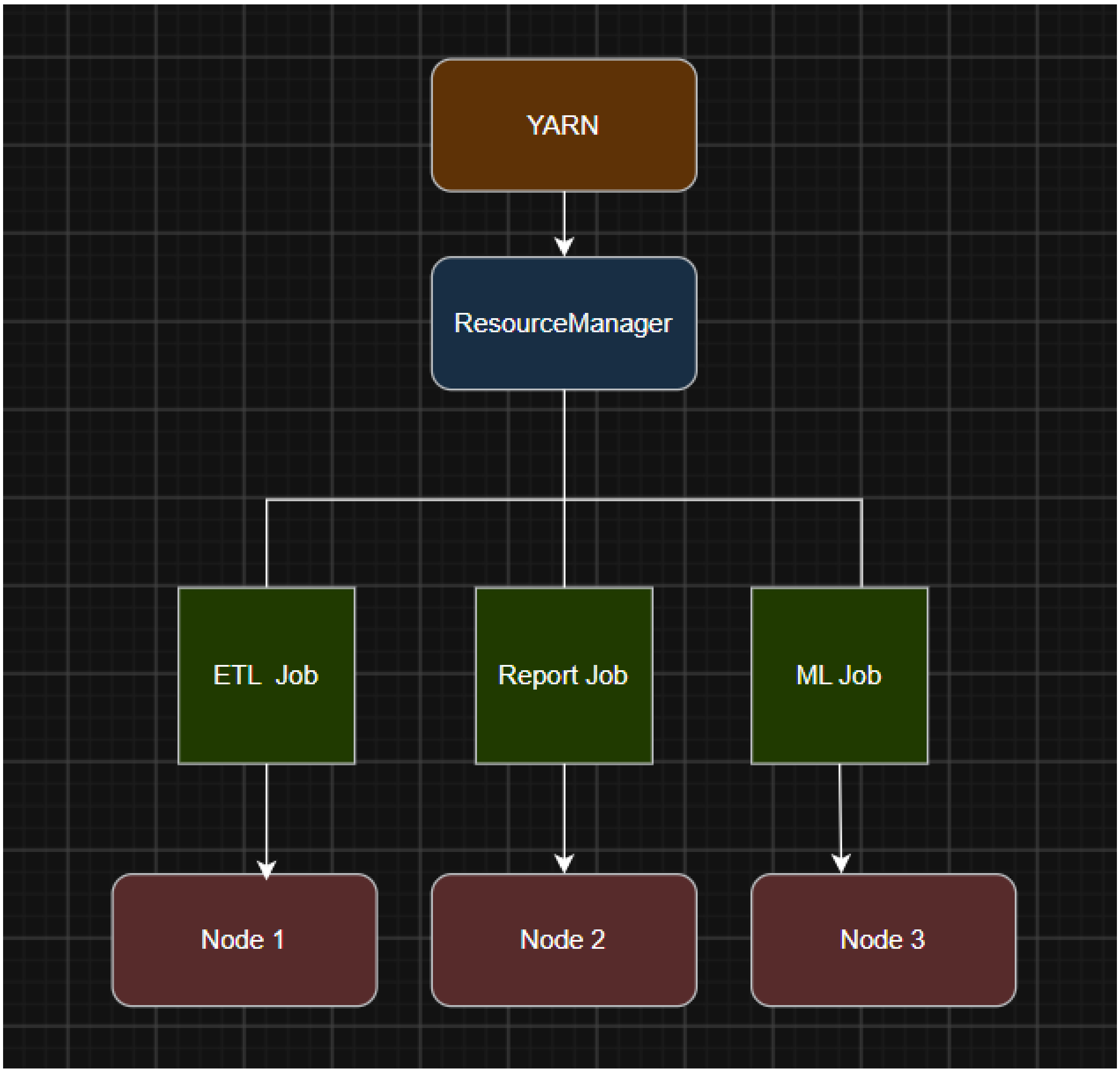
4. What role does YARN play in large-scale Hadoop processing?

Ans:

YARN stands for Yet Another Resource Negotiator. Its main role in Hadoop is to manage cluster resources and schedule jobs.

YARN has a ResourceManager that manages the overall resources and NodeManagers that manage resources on individual worker nodes.

When multiple jobs are submitted, YARN decides how much CPU and memory each job should receive. This allows different applications to run on the same Hadoop cluster.



For example, if an ETL job and a machine learning job are submitted at the same time, YARN allocates resources to both according to the scheduling policy.

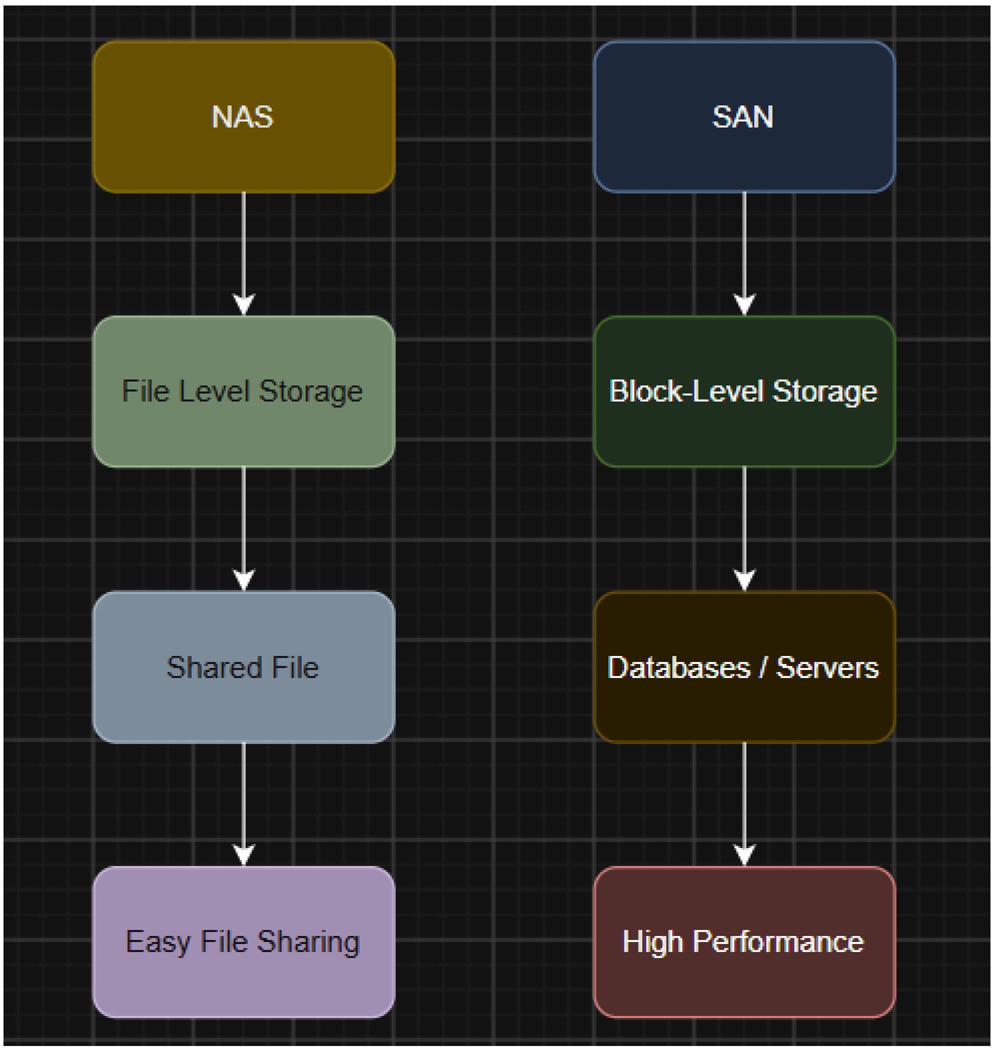
5. Differentiate NAS and SAN in an enterprise data environment.

Ans:

NAS and SAN are both used for storing data in an enterprise, but they work differently.

NAS (Network Attached Storage) provides file-level storage over a network. It is mainly used for sharing files such as documents, images and videos between users and systems.

SAN (Storage Area Network) provides block-level storage. It gives servers direct access to storage blocks and is commonly used for high-performance applications and databases**.**



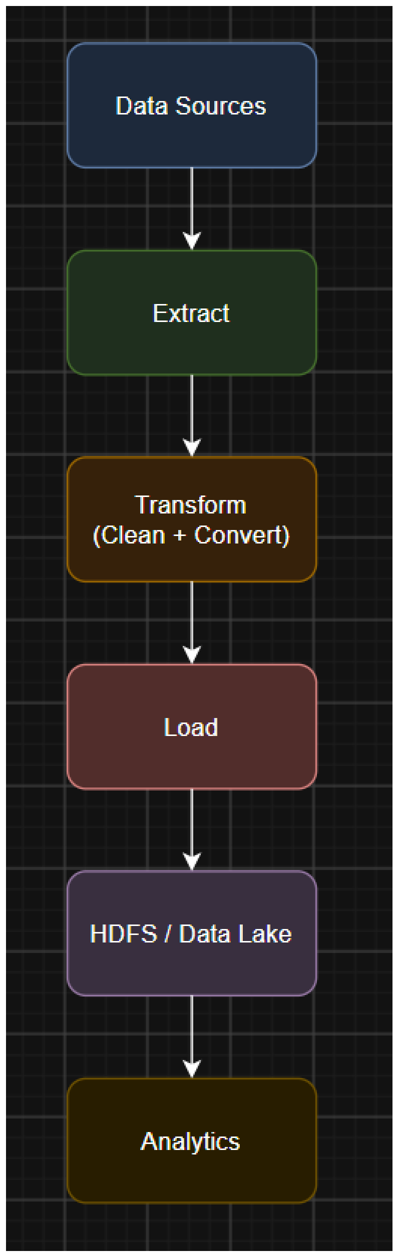
For example, an organization can use NAS for sharing employee documents, while SAN can be used for a high-performance database.

6. What is the purpose of ETL in a big data pipeline?

Ans:

ETL stands for Extract, Transform and Load. Its purpose is to collect data from different sources, clean and convert it, and then store it in a big data system for analysis.

- Extract: Collect data from databases, files, applications, sensors, etc.
- Transform: Clean the data, remove duplicates, handle missing values and convert it into a common format.
- Load: Store the processed data in systems such as HDFS, a data lake or data warehouse.



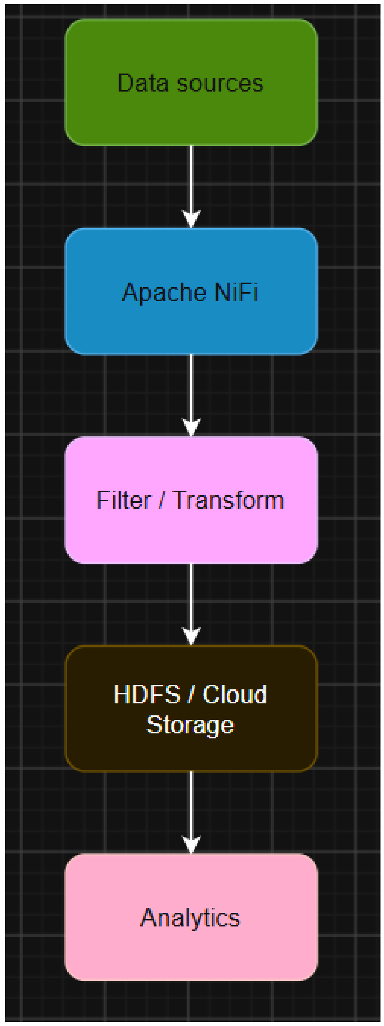
For example, a retail company can collect customer data from different stores, remove duplicate records and convert the data into a common format before storing it for sales analysis.

7. How does Apache NiFi help in data integration workflows?

Ans:

Apache NiFi is a **data integration tool** used to move and manage data between different systems. It can collect data from databases, files, applications, APIs and other sources and send it to systems such as HDFS or cloud storage.

NiFi can also **filter, transform, route and monitor** data while it is moving through the pipeline.



For example, a company can use NiFi to collect customer data from different databases, remove unwanted records and send the required data to HDFS for analysis.

Another advantage is that NiFi provides a **visual flow interface**, making it easier to monitor data movement and handle errors.

8. What are the key failure points in a Hadoop cluster and how would you handle them?

Ans:

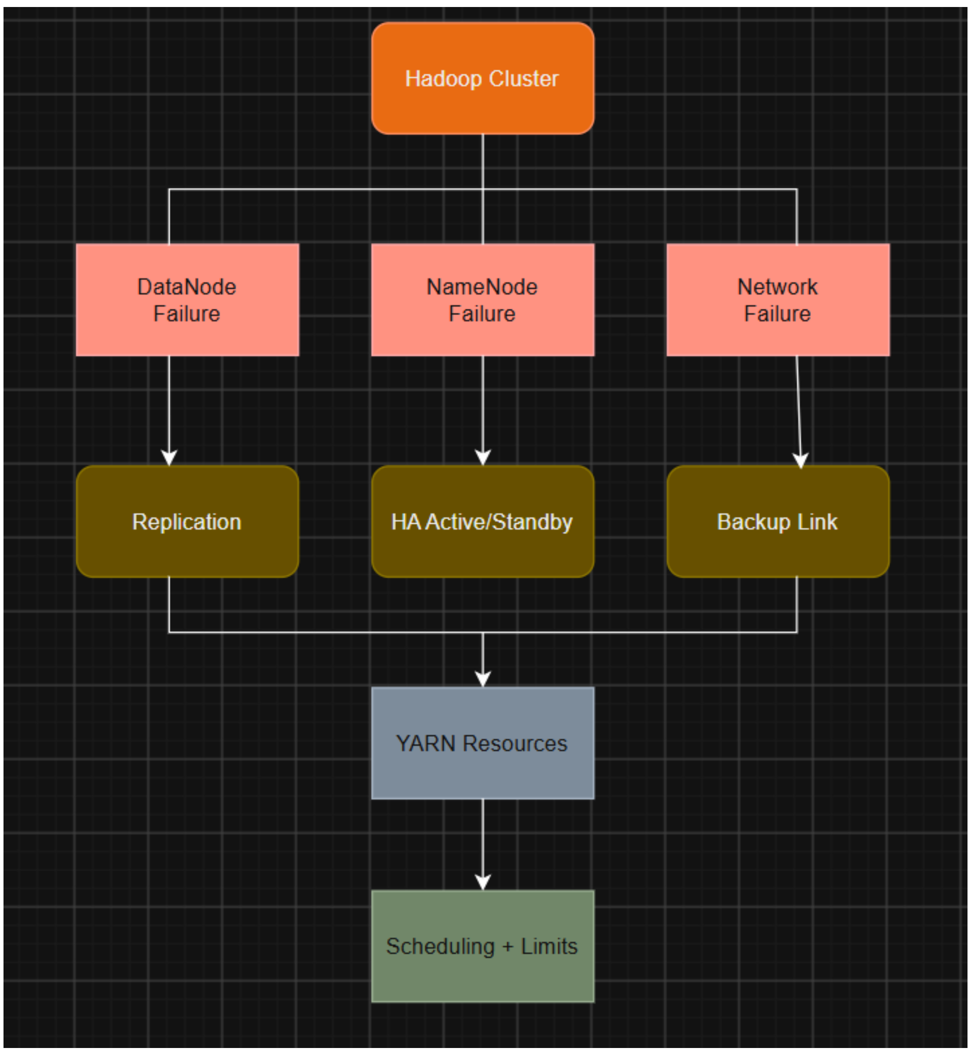
The main failure points in a Hadoop cluster are DataNodes, NameNode, network, and YARN resources.

If a DataNode fails, HDFS replication provides another copy of the data. The failed node can later be replaced and the missing replica can be recreated.

If the NameNode fails, the Hadoop system may not be able to access the file metadata. This can be handled using NameNode High Availability, with an Active and Standby NameNode.

Network failure can affect communication between nodes. Redundant network connections and proper monitoring can help reduce this problem.

YARN resource problems can occur when one job uses too much CPU or memory. This can be handled using queues, resource limits and proper scheduling.



For example, if one DataNode containing a file block fails, HDFS can use another replica instead of making the data unavailable.

9. How would you design a secure data ingestion pipeline for a large organization?

Ans:

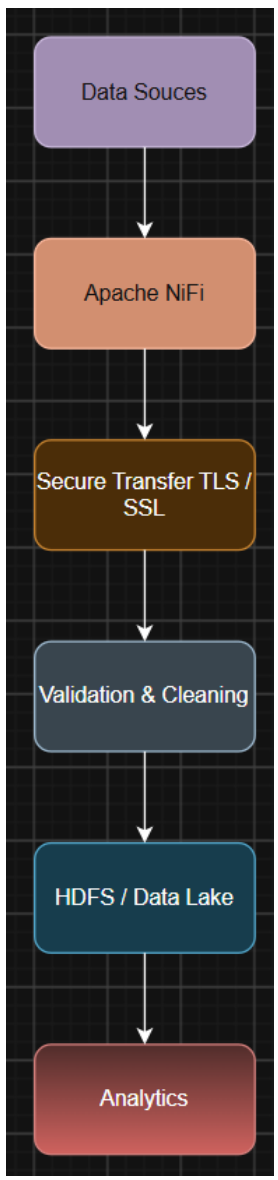
I would design the pipeline so that data is collected, secured, validated and stored before it is used for analysis.

First, data can be collected from databases, applications, files and APIs using a tool such as Apache NiFi. During data transfer, TLS/SSL encryption should be used so that the data cannot be easily accessed by unauthorized people.

Authentication should be used to verify the source of the data, and Role-Based Access Control (RBAC) should be used to give access only to authorized users.

After receiving the data, it should be validated and cleaned. Duplicate or invalid records can be removed before storing the data in HDFS or a cloud data lake. Sensitive data should also be encrypted while stored.

A simple architecture is:



The system should also maintain **audit logs** to record data access and important activities. Monitoring can be used to detect failed transfers or suspicious activity.

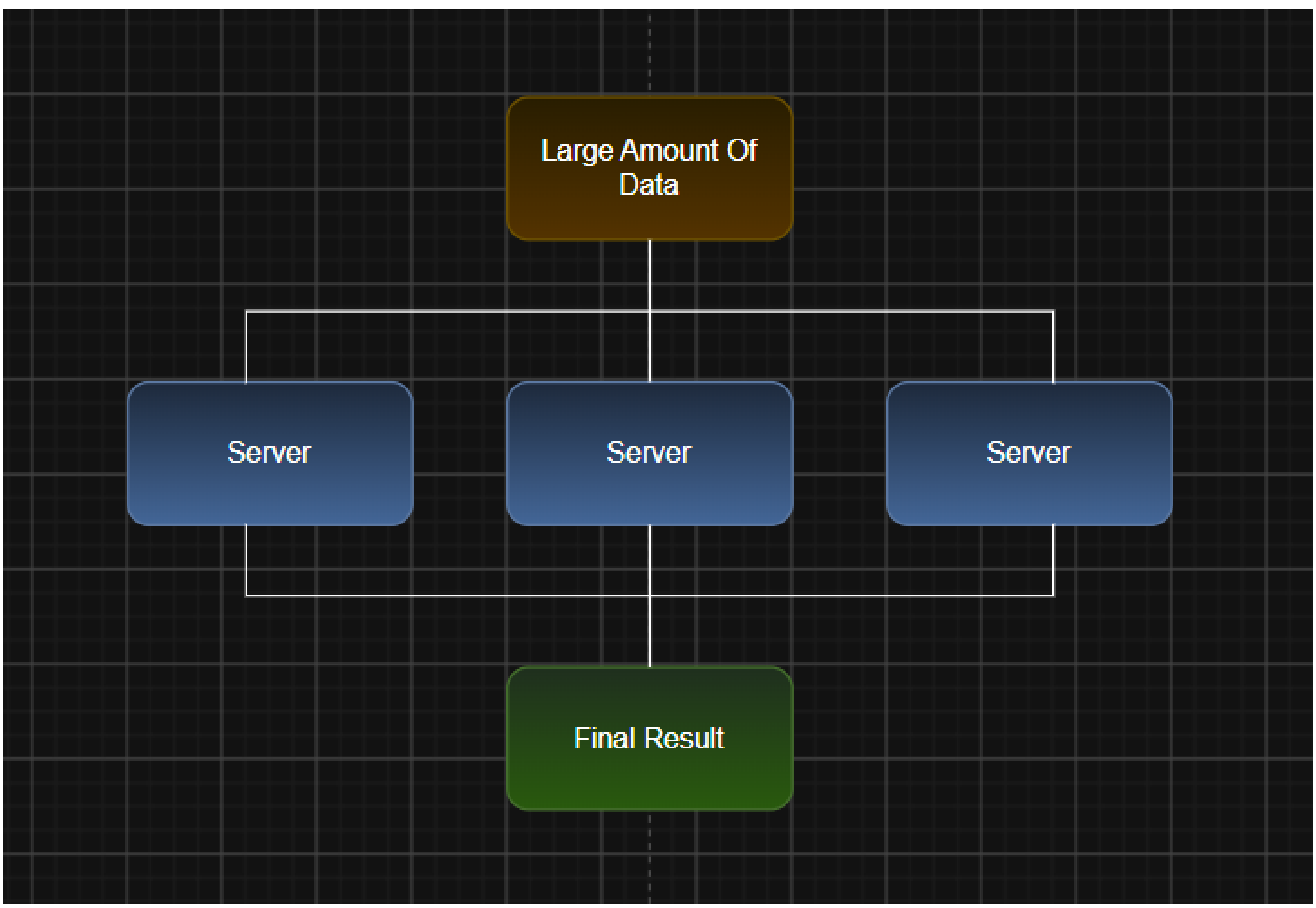
For example, a bank can use this pipeline to securely transfer transaction data from its banking systems to a Hadoop platform for fraud analysis.

10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

Ans:

I would explain it with a simple example. Imagine a company has **one person** who has to process millions of files. It would take a very long time. Instead, if we give the work to many people and let them work at the same time, the work can be completed much faster. Hadoop works in a similar way.

In Hadoop, **HDFS** stores large amounts of data across many computers. **MapReduce or Spark** divides the processing work among these computers, and **YARN** manages the available CPU and memory resources.



The main benefit is **scalability**. If the company's data increases, we can add more computers to the Hadoop cluster instead of replacing the whole system.

For example, an e-commerce company can use Hadoop to process millions of customer transactions and generate sales reports.

So, in simple words, **Hadoop makes many computers work together as one large processing system**, allowing the company to handle increasing amounts of data efficiently.